

Charger (BQ24250): Working Document

Digital Audio Player • Rev 9

Author: Alexander Loker. Design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic). Structure and prose from the author's direction and source data.

Contents

1	Summary	1
2	Design Requirements	1
3	Pin Configuration	1
4	External Components	2
5	Equations and Results	3
6	Important Values	4
7	Open Problems	4
8	Revision Log	4
9	Sources	5

1. Summary

Field	Value
Manufacturer / MPN	Texas Instruments, BQ24250RGET (AL-PN #1-054)
Package	24-VQFN, 4 × 4 mm
Function	3 MHz switch-mode single-cell Li-ion charger with PowerPath (SYS output)

2. Design Requirements

Requirement	Value
Input	USB-C, 5 V
Battery	1S Li-Po, 3.0 to 4.2 V, 1850 mAh
Charge current	500 mA (0.27C)
Input current limit	1 A
V_{IN_DPM}	4.488 V
Switching frequency	3 MHz (2.7 to 3.3 MHz)
Charge temperature window	TS network, 0 to 60 °C
System peak (S4)	403 mA at 3.7 V (Power Budget Rev 6)
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1	CE	Default low (charging enabled); firmware can disable
2	EN1	10 k Ω pull-up to +3V3_DIG
3	EN2	10 k Ω pull-down to GND
4	AGND	GND
5	SDA	10 k Ω pull-up to +3V3_DIG
6	SCL	10 k Ω pull-up to +3V3_DIG
7	STAT	Charge LED (green) and 1 k Ω from SYS
8	INT	10 k Ω pull-up to +3V3_DIG
9	TS	R2 from LDO; R3 and NTC to AGND
10	ISET	R_{ISET} to AGND
11, 12	BAT	Battery +, C_{BAT}
13, 14	SYS	System rail, C_{SYS} bank
15, 16	PGND	GND
17, 18	SW	Inductor, C_{BOOT}
19	IN	VUSB (filtered), C_{IN} , V_{DPM} divider
20	PMID	C_{PMID} to GND
21	BOOT	C_{BOOT} to SW
22	ILIM	R_{ILIM} to AGND
23	VDPM	Divider tap
24	LDO	C_{LDO} , top of TS network
25	Thermal pad	GND

4. External Components

Ref	Value / Rating	MPN	AL-PN
L	1.0 μH $\pm 20\%$, 2 A I_{sat} , 96 m Ω , 0505	TDK TFM141206BLE-1R0MTCA	#1-063
C_{BAT} , C_{LDO} , C_{PMID}	1 μF 16 V X6S 0402	Murata GRM155C81C105KE11D	#1-058
C_{BOOT}	33 nF 16 V X7R 0402	Murata GRM155R71C333KA01D	#1-059
C_{IN} (C3)	2.2 μF 25 V X7R 0603	Murata GRM188Z71E225KE43D	#1-070
C_{SYS1} , C_{SYS2}	22 μF 25 V X7S 1210	Murata GCM32EC71E226KE36L	#1-073
C_{SYS3}	0.1 μF 50 V X7R 0402	Murata GRM155R71H104KE14D	#1-048
R_{ISET}	500 Ω 0.1% 0603	YAGEO RT0603BRC07500RL	#1-064
R_{ILIM}	270 Ω 1% 0603	YAGEO RC0603FR-07270RL	#1-065
R_{DPM1} / R_{DPM2}	274 k Ω / 100 k Ω 1% 0603	Generic	
R2 (TS)	5 k Ω 1% 0402	YAGEO RC0402FR-075KL	#1-060
R3 (TS)	9.88 k Ω 0.5% 0402	YAGEO RT0402DRD079K88L	#1-061
NTC	10 k Ω , β 4000 K, on cell body		

Ref	Value / Rating	MPN	AL-PN
F1	120 Ω at 100 MHz, 3 A, 0805	Würth 742792023	#1-068
C1, C2 (filter)	4.7 μ F 16 V X7R 0805	Murata GRM21BR71C475KE51L	#1-071
R1 (filter damping)	0.5 Ω 1% 1/4 W 0805, pulse-rated metal film	YAGEO PE0805FRF7W0R5L	#1-097
D1	SMAJ7.0A TVS, SMA	Diodes Inc. SMAJ7.0A-13-F	#1-049
D2, D3	2-ch ESD, 5.5 V, 1.5 pF, SC-70-3 (D2: D+/D-, D3: CC1/CC2)	TI TPD2E2U06QDCKRQ1	#1-099
J1	USB-C receptacle, USB 2.0	GCT USB4085-GF-A	#1-069
J	Battery connector, 3-pos, 5.08 mm TH	Phoenix Contact 1786417	#1-062
R_{CC1}, R_{CC2}	5.1 k Ω 0402	Generic	
Pull-ups / pull-down	10 k Ω 0603 ($\times 5$ / $\times 1$)	Generic	

5. Equations and Results

Inductor ripple ($V_{IN} = 5$ V, $V_{BAT} = 3.0$ V, $f_s = 2.7$ MHz, $L = 1.0$ μ H)

$$\Delta I_L = \frac{V_{BAT} (V_{IN} - V_{BAT})}{V_{IN} f_s L} = 0.444 \text{ A}$$

$$I_{PEAK} = I_{Load} + \frac{\Delta I_L}{2}, \quad I_{Load} = I_{SYS} + I_{CHG}$$

Source: [bq2425x Product Page and Datasheet \(Texas Instruments\)](#), inductor selection.

Case	I_{avg}	I_{PEAK}
Charging, idle	0.500 A	0.722 A
Charging + S2 playback	0.664 A	0.886 A
Charging + S4 peak	0.903 A	1.125 A (vs 2 A I_{sat})

Charge current, input current limit and input DPM threshold

$$R_{ISET} = \frac{250}{I_{FC}} = \frac{250}{0.5} = 500 \Omega$$

$$R_{ILIM} = \frac{270}{I_{IC}} = \frac{270}{1.0} = 270 \Omega$$

$$V_{IN_DPM} = 1.2 \frac{R_{DPM1} + R_{DPM2}}{R_{DPM2}} = 1.2 \times \frac{374}{100} = 4.488 \text{ V}$$

Source: [bq2425x Product Page and Datasheet \(Texas Instruments\)](#).

TS thresholds are ratiometric to V_{LDO} (10 k Ω , β 4000 K NTC):

Threshold	V_{TS} / V_{LDO}	Temperature
Cold	60%	0 °C
Cool	56.5%	≈ 13 °C
Warm	38.3%	≈ 47 °C
Hot	30%	60 °C

Input current limit straps (bq2425x Table 1):

EN2	EN1	Input limit	V_{IN_DPM}
0	1	External R_{ILIM} (1 A)	External divider (4.488 V); operating state
0	0	500 mA	4.36 V; default while +3V3_DIG is down

Filter damping resistor pulse check (hot plug, C = 4.7 μF)

$$E = \frac{1}{2} CV^2 = \frac{1}{2} \cdot 4.7 \mu\text{F} \cdot (5 \text{ V})^2 = 59 \mu\text{J}, \quad \tau = 0.5 \Omega \cdot 4.7 \mu\text{F} = 2.4 \mu\text{s}, \quad P_{pk} \leq \frac{(5 \text{ V})^2}{0.5 \Omega} = 50 \text{ W}$$

Source: PE series 0805 pulse rating (≈ 200 W single pulse at 100 μs), giving ≥ 4× margin.

6. Important Values

Item	Value
Input filter insertion loss	–83 dB at 3 MHz
Filter output impedance peak	0.46 Ω with damping (46 Ω without); limit 6.25 Ω
SYS ripple (sim)	2.4 mV pp
Inductor DCR loss (charging + playback)	≈ 42 mW
Inductor peak, worst case	1.125 A (56% of 2 A I_{sat})
Battery FET R_{ON} (BAT to SYS)	30 mΩ typ, 40 mΩ max
IC current limit	2.6 A min
VBUS capacitance	11.6 μF nominal; < 10 μF effective at 5 V, within the USB 2.0 limit
Grounding	AGND and GND on a single ground

7. Open Problems

#	Problem
1	I2C may only work while USB is present (powered from IN); verify before writing the driver.
2	NTC return shares J200 pin 3 with the battery current (0.5 A while charging), which offsets the TS reading by well under 1 °C. Attach the NTC return lead at the connector end of the harness (pin 3 contact, not the cell tab) and give pin 3 a short, wide path into the ground plane near the charger AGND.
3	Battery connector size.

8. Revision Log

Rev	Date	Change
0	2026-09-19	Initial issue: part selection, sizing, calculations, TS network, input filter and SYS ripple simulations.
1	2026-09-19	Reviewed against schematic; pin connections added.
2	2026-09-21	Constraints added; cell charge-temperature mismatch noted; condensed.
3	2026-09-22	S4 system peak 462 → 385 mA; inductor peak 1.107 A.
4	2026-09-23	R1 to pulse-rated PE 0805; USB D+/D- ESD; EN1/EN2 strap verified.
5	2026-09-23	Second TPD2E2U06 on CC1/CC2.
6	2026-09-23	Single ground; VBUS capacitance confirmed.
7	2026-09-25	Reformatted; linked contents and sources; open problems 4 and 5 removed; connector item reworded.
8	2026-09-26	NTC return open problem reworded to the harness and routing mitigation.
9	2026-09-27	System load from Power Budget Rev 6: S4 peak 403 mA, inductor peak 1.125 A.

9. Sources

1. [bq2425x Product Page and Datasheet \(Texas Instruments\)](#)
2. [TPD2E2U06-Q1 Product Page and Datasheet \(Texas Instruments\)](#)